

The propositions package*

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Abstract

The `propositions` package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like ‘(P)’ or ‘Physicalism’, or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with `amsmath`, `hyperref`, and `cleveref`.

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1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises, . . .) with various kinds of labels. A thesis might be referred to as ‘(P)’ or ‘Physicalism’; the premises of an argument might be numbered as ‘P1’, ‘P2’, ‘P3’, . . . ; or examples might be numbered consecutively over the course of a whole article. Standard L^AT_EX environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The `propositions` package solves this by attaching formatting information to each label. A short item like `\item[P]` (inside `prop`) is displayed as “(P)” and `\ref` automatically produces “(P)” as well—complete with parentheses, hyperlinks, and `cleveref` support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires `amsmath`) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in L^AT_EX3, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[options]{propositions}` (see [section 8](#) below for valid package options).

The `prop` environment generates a list of propositions, each introduced by `\item`. `\item` with an optional argument gives a `description`-like label:

<pre>\begin{prop} \item[Physicalism] Everything is physical. \label{phys} \item[Idealism] Everything is mental. \label{ideal} \end{prop}</pre>	<pre>Physicalism Everything is physi- cal. Idealism Everything is mental.</pre>
--	---

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

<pre>\ref{phys} is more plausible than \ref{ideal}.</pre>	<pre>Physicalism is more plausible than Idealism.</pre>
---	---

With no optional argument, `\item` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

<pre> \begin{prop} \item Every atom is physical. \label{atoms} \end{prop} \ref{phys} follows from the conjunction of \ref{atoms} and \begin{prop} \item Everything is an atom. \label{atomism} \end{prop} </pre>	(1) Every atom is physical. Physicalism follows from the conjunction of (1) and (2) Everything is an atom.
--	--

As with `enumerate`, the counter and formatting depend on the nesting level:

<pre> \begin{prop} \item \label{dual} \begin{prop} \item Some things are physical. \label{some} \item Some things are not physical. \label{notall} \end{prop} \end{prop} \end{prop} Without \ref{some}, \ref{dual} would be consistent with \ref{ideal}. </pre>	(3) a. Some things are physical. b. Some things are not physical. Without (3a), (3) would be consistent with Idealism.
---	--

4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see [section 5](#) for the possible keys):

<pre> \begin{prop} \item[No Overlap, align=flush, display format=\textsc{#1}, ref format=\textit{#1}] Nothing mental is physical. \label{incomp} \end{prop} Is \ref{incomp} consistent with \ref{phys}? </pre>	No OVERLAP Nothing mental is physical. Is <i>No Overlap</i> consistent with Physicalism?
--	---

Preset *prop styles* can be declared and used instead of configuring the `\items` one at a time:

<pre> \begin{prop} \item[Nihilism, style=thesis] There is nothing. \label{nihilism} \end{prop} Does \ref{nihilism} imply \ref{phys}, \ref{dual}, or both? Discuss. </pre>	NIHILISM There is nothing. Does NIHILISM imply Physicalism, (3), or both? Discuss.
---	---

Shortcut commands—e.g., `\titem[⟨keys⟩]` for `\item[style=thesis, ⟨keys⟩]` (within `prop`)—can also be defined. The package loads with a range of predefined styles.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\items` within `prop` use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

<pre>\begin{equation} \exists x (\text{Mental}(x) \wedge \text{Physical}(x))</pre>	$(4) \quad \exists x(\text{Mental}(x) \wedge \text{Physical}(x))$
<pre>\end{equation} \begin{prop} \item There is overlap between the mental and the physical. \end{prop}</pre>	$(5) \quad \text{There is overlap between the mental and the physical.}$

The `\ptag` command (requires `amsmath`) is an analogue of `\item` (within `prop`) that works inside displayed math environments.

<pre>\begin{equation} \ptag{Monism} \label{mon} \exists x \forall y (y = x)</pre>	$\text{Monism} \quad \exists x \forall y (y = x)$
<pre>\end{equation} Is \ref{mon} compatible with \ref{dual}?</pre>	$\text{Is Monism compatible with (3)?}$

5 The `prop` environment and `\pitem`

```
\begin{prop}[⟨keys⟩]
  ⟨environment content⟩
\end{prop}
```

Creates a displayed list of propositions. It is a standard L^AT_EX list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting package keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper. The optional `⟨keys⟩` argument accepts the same keys as `\proppositions`^{P. 13} (section 8), with effects local to this environment.

```
\begin{inlineprop}[⟨keys⟩]
  ⟨environment content⟩
\end{inlineprop}
```

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional `⟨keys⟩` as `prop`.

Within `prop` and `inlineprop`, `\item` is redefined to act as the proposition-item command. Its optional argument is a comma-separated list of $\langle key \rangle = \langle value \rangle$ pairs (the keys are described below). A bare string without `=` is treated as a proposition name.

When used without an optional argument (or without setting `name`, `counter`, or `style`), the style is determined by the `nameless style` key (section 8). By default this is `numbered`, which dispatches by nesting depth to `levelone`, `leveltwo`, ... via `\proplevelchoice`.

`\pitem[ $\langle keys \rangle$ ]` (deprecated)

Deprecated alias for `\item` within `prop` or `inlineprop`. Issues a warning on each use. Replace with `\item`.

`\ptag[ $\langle keys \rangle$ ]`

(Available only when `amsmath` is loaded.) Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\item` within `prop`, except that `align` has no effect (since positioning is controlled by the tag placement system).

The following keys can be used in the optional argument of `\item` (within `prop` or `inlineprop`) or the argument of `\ptag`:

`name= $\langle text \rangle$` (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name= $\langle text \rangle$` .

`style= $\langle style \rangle$` (no default)

A prop style, equivalent to a preset collection of keys. Styles can be declared using `\SetPropStyle`^{P.7} or `\DeclareNumberedStyle`^{P.7}, and several come predefined (section 6).

`counter= $\langle name \rangle$` (no default)

Counter to use. The counter is automatically stepped, and the item's `name` is set to `\the $\langle name \rangle$` (though this can be overridden by `counter format` or `name`).

`counter format= $\langle template \rangle$` (no default)

How to display the counter value. Use `#1` for the counter name, e.g. `counter format=\roman{#1}`. When set, this is used instead of `\the $\langle counter \rangle$` for generating the item's name.

`format= $\langle template \rangle$` (no default)

Formatting applied to the `name`: use `#1` for the argument, e.g. `format=\textbf{(#1)}`. Shorthand for setting both `display format` and `ref format`.

`display format= $\langle template \rangle$` (no default)

Format for displaying the name in the proposition's label. Does not affect cross-references.

`label format= $\langle template \rangle$` (no default)

Format applied to the *entire* assembled label, i.e. the name (after `display format`), shorthand, and gloss together. Use `#1` for the whole assembled text. For example, `label format=#1`: appends a colon after the complete label. Does not affect cross-references.

ref format= $\langle template \rangle$ (no default)
Format for subsequent cross-references to this proposition. Does not affect the display.

shorthand= $\langle text \rangle$ (no default)
An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with **ref format**) rather than the full name.

shorthand format= $\langle template \rangle$ (initially $\sim[\#1]$)
Format for displaying the shorthand in the label.

gloss= $\langle text \rangle$ (no default)
A parenthetical gloss displayed after the name. Does not affect cross-references.

gloss format= $\langle template \rangle$ (initially $\sim(\#1)$)
Format for displaying the gloss in the label.

ref= $\langle text \rangle$ (no default)
Explicitly set the reference text, overriding what would be derived from **name**, **counter**, or **shorthand**.

label= $\langle label \rangle$ (no default)
Equivalent to a trailing `\label{\langle label \rangle}`.

crefname= $\langle type \rangle$ (no default)
When **cleveref** is loaded, assigns an arbitrary reference type to this proposition. For example, **crefname=lemma** causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default **proposition** type. The $\langle type \rangle$ must be known to **cleveref**; new types can be declared with `\crefname`.

reset= $\langle boolean \rangle$ (initially **true**)
When **true** (the default), processing this `\pitem` resets the sub-level counter one nesting depth below the current level (e.g. **numpropii** at level 1). Set **reset=false** to suppress this reset, so that sub-item numbering continues from where it left off across consecutive parent items. Can also be set in a style definition via `\SetPropStyle`^{P.7}.

align= $\langle type \rangle$ (no default)
How the label should be positioned. Possible values: **left** (standard left-aligned label, offset controlled by `\labelwidth` and `\labelsep`), **right** (right-aligned, like **enumerate**), **center** (centered within the label box), **flush** (aligned with left margin of item text), **nextline** (label on its own line), and **flush-nextline**. Has no effect inside `inlineprop` or `\ptag`. For left-margin alignment, use `labelindent=0em` instead.

labelsep= $\langle length \rangle$
itemindent= $\langle length \rangle$
listparindent= $\langle length \rangle$
labelindent= $\langle length \rangle$

These keys override the settings of the corresponding list dimensions just for the relevant `\pitem`. For the role of `labelindent` (which determines the distance from enclosing left margin to the left edge of the label box), see the corresponding package option.

6 Prop styles

`\SetPropStyle{<name>}{<keys>}`

Defines or modifies a prop style for use with the `style` key. All `\pitem`^{P.5} keys are accepted, plus the following:

`style=<parent>` (no default)

Inherit from a parent style. When an item is created, the parent's settings are loaded first (recursively, if the parent itself has a parent), then this style's own keys are applied on top.

The argument may be a macro which is expanded when the item is created: for example, a style with `style=\{<style1>\},\{<style2>\},\{<style3>\}` will resolve to `\{<style1>\}, \{<style2>\}, \{<style2>\}` depending on the nesting depth. The built-in `numbered` and `eqnum` styles use this mechanism.

`macro=<command>` (no default)

A new user macro, equivalent to `\pitem[style=<name>]`. Any further keys given to the macro are passed to `\pitem`.

If the style `<name>` already exists, `\SetPropStyle` modifies or adds keys. For example, `\SetPropStyle{proposition}{align=flush}` changes the alignment of the built-in `proposition` style while preserving its other settings.

Styles can also be created with a `pgfkeys`-like syntax via `\propoptions`^{P.13}: `\propoptions{<name>/ .style={<keys>}}` is equivalent to `\SetPropStyle{<name>}{<keys>}`.

<pre> \SetPropStyle{angle}{ labelindent = 0em, display format = \textbf{<\$\langle\$#1\$\rangle\$}, ref format = <\$\langle\$#1\$\rangle\$, macro = \angitem } \begin{prop} \angitem[Angle thesis] Everything is angular. \end{prop} No further discussion of \lastref{} is needed. </pre>	<code><Angle thesis></code> Everything is angular. No further discussion of <code><Angle thesis></code> is needed.
--	---

`\DeclareNumberedStyle{<name>}[<keys>]`

Creates a new L^AT_EX counter named `<name>` and a matching prop style with `counter=<name>`. All `\SetPropStyle` keys are accepted, plus:

`parent=<counter>` (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as `\numberwithin`). A dedicated `prop` counter (stepped by each `prop` and `inlineprop`) is available for non-persistent numbering.

<pre> \DeclareNumberedStyle{P} \begin{prop} \item[counter=P] First premise. \label{p1} \item[counter=P] Second premise. \label{p2} \end{prop} From \ref{p1} and \ref{p2}\ldots </pre>	P1 First premise. P2 Second premise. From P1 and P2...
---	---

`\proplevelchoice{<item1, item2, ...>}`

Picks the entry at the current nesting depth (1-indexed, clamped to the list length). At depth 0 (outside any `prop` environment), returns the first entry. This macro is fully expandable and can be used in the `name`, `format`, and `style` keys of `\SetPropStyle` to create depth-dependent styles.

6.1 Built-in styles

The following prop styles are predefined. Each entry shows the defining code (using user-facing commands) and a live example. All styles can be modified with `\SetPropStyle`^{P. 7}.

Text styles

`plain` Unformatted text label.

<pre> \SetPropStyle{plain}{format = #1} </pre>	Claim An unformatted item.
--	-----------------------------------

`proposition` Bold text label. Auto-selected when `\item` has a name but no style.

<pre> \SetPropStyle{proposition}{format = \textbf{#1}} </pre>	Thesis A bold-formatted item.
---	---

`thesis` Small-caps name at the text margin (flush alignment). Shortcut: `\titem`.

<pre> \SetPropStyle{thesis}{format = \textsc{#1}, align = flush, macro = \titem} </pre>	HYPOTHESIS A flush-aligned item.
---	---

`vignette` Italic name at the text margin with a colon in the label.


```
| \SetPropStyle{vignette}{ref format = \textit{#1},
| align = flush, label format = \textit{#1:}}
```

Example: A vignette item.

bullet Bullet symbol varying by depth (like `\itemize`). Shortcut: `\bitem`.

```
| \SetPropStyle{bullet}{align = center,
| name = \proplevelchoice{\textbullet,
| {\normalfont\textendash}, \textasteriskcentered,
| \textperiodcentered},
| display format = #1, macro = \bitem}
```

- A bullet item.

Numbered styles with dedicated counters

roman Roman-numeral counter, resets per section. Shortcut: `\ritem`.

```
| \DeclareNumberedStyle{roman}[parent = section,
| counter format = \roman{#1}, format = (#1), macro = \ritem]
```

- (i) A roman-numbered item.

alph Alphabetic counter, resets per section. Shortcut: `\aitem`.

```
| \DeclareNumberedStyle{alph}[parent = section,
| counter format = \alph{#1}, format = (#1), macro = \aitem]
```

- (a) An alphabetic item.

Level-specific styles

These styles are used by the **numbered** and **eqnum** dispatcher styles (below). They share the counters `numpropi`–`numpropv`, which reset automatically when an `\item` at the next lower level is processed.

levelone Level-1 numbered items: arabic in parentheses.

```
| \SetPropStyle{levelone}{counter = numpropi, format = (#1)}
```

leveltwo Level-2: lowercase alphabetic with dot.

```
| \SetPropStyle{leveltwo}{counter = numpropii,
| display format = #1., ref format = \parentref{#1}}
```

levelthree Level-3: lowercase roman in parentheses.

```
| \SetPropStyle{levelthree}{counter = numpropiii,
| display format = (#1), ref format = \parentref{.#1}}
```

levelfour Level-4: uppercase alphabetic with dot.

```
| \SetPropStyle{levelfour}{counter = numpropiv,  
|   display format = #1., ref format = \parentref{#1}}
```

levelfive Level-5: uppercase roman in parentheses.

```
| \SetPropStyle{levelfive}{counter = numpropv,  
|   display format = (#1), ref format = \parentref{.#1}}
```

equation Uses the `equation` counter. When the `equations` option is active, `\SetPropStyle{equation}{...}` automatically updates the `\tagform@` and `\ref` hooks for all equations.

```
| \SetPropStyle{equation}{counter = equation, format = (#1)}
```

Dispatcher styles

These styles use `style` with `\proplevelchoice` to dispatch to a level-specific style at resolution time.

numbered The default nameless style. Dispatches to `levelone`–`levelfive` by nesting depth.

```
| \SetPropStyle{numbered}{style = \proplevelchoice{  
|   levelone, leveltwo, levelthree, levelfour, levelfive}}
```

eqnum Like `numbered`, but uses `equation` at level 1. Activated by the `equations` package option (which sets `nameless style = eqnum`).

```
| \SetPropStyle{eqnum}{style = \proplevelchoice{  
|   equation, leveltwo, levelthree, levelfour, levelfive}}
```

hierarchical Produces 1, 1.1, 1.1.1... numbering using `\parentref` and `counter format`. Defined via two helper styles:

```
| \SetPropStyle{h-base}{counter = numpropi,  
|   counter format = \arabic{#1}, format = #1}  
| \SetPropStyle{h-sub}{  
|   counter = \proplevelchoice{numpropi, numpropii,  
|     numpropiii, numpropiv, numpropv},  
|   counter format = \arabic{#1},  
|   format = \parentref{.#1}}  
| \SetPropStyle{hierarchical}{style = \proplevelchoice{  
|   h-base, h-sub}}
```

7 Cross-referencing

Labels placed after `\item` items (within `prop`) work with the standard `\label/\ref` mechanism. The key difference from ordinary L^AT_EX references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

`\Ref{<label>}`
`\Ref*{<label>}`

Titlecase variants of `\ref` and `\ref*`: uppercases the first letter of the formatted output. (For example, if `\ref{thesis}` produces ‘the Identity Theory’, then `\Ref{thesis}` produces ‘The Identity Theory’.) The starred form suppresses the hyperlink. Note: `\Ref` only affects references produced by this package (which are stored via `\propapply`); for other references, it behaves like `\ref`.

`\nref{<label>}`
`\nref*{<label>}`
`\Nref{<label>}`
`\Nref*{<label>}`

“Naked ref.” Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces ‘(P1)’, then `\nref{premise}` produces ‘P1’. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` can be useful in the argument of `\item`, when the the name of one proposition should depend on that of another:

<code>\SetPropStyle{proposition}{format=(#1)}</code>	
<code>\begin{prop}</code>	(Phys) Everything is physical.
<code>\item[Phys]</code>	(Phys*) Almost everything is physical.
Everything is physical.	ical.
<code>\label{phys2}</code>	
<code>\item[\nref{phys2}*]</code>	((Phys*)) This one has two sets
Almost everything is physical.	of parentheses, which is probably
<code>\label{newphys2}</code>	not desired. (Phys*) avoided this
<code>\item[\ref{phys2}*]</code>	by using <code>\nref</code> .
This one has two sets of parentheses,	
which is probably not desired.	
<code>\ref{newphys2}</code> avoided this by using	
<code> \nref </code> .	
<code>\end{prop}</code>	

Warning: documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple L^AT_EX runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

`\oref [<prefix>] [<suffix>]{<label>}`
`\oref* [<prefix>] [<suffix>]{<label>}`
`\Oref [<prefix>] [<suffix>]{<label>}`
`\Oref* [<prefix>] [<suffix>]{<label>}`

“Ref with options.” Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, $\langle suffix \rangle$ is appended; with two, $\langle prefix \rangle$ is prepended and $\langle suffix \rangle$ appended. For instance, if `\ref{premise}` produces ‘(P1)’, then `\oref[*]{premise}` produces ‘(P1*)’ and `\oref[old~][*]{premise}` produces ‘(old~P1*)’.

`\Oref` is the titlecase variant; the starred forms suppress the hyperlink.

`\oref` can also be useful in the name of `\items`, if one wants the display format for the modified item to depend on that originally used

```
\begin{prop}
  \label{premise}
  \item[style=plain,name=\oref[*]{phys2}]
    This will use boldface and
    parentheses because the original
    referenced item did.
\end{prop}
```

(Phys*) This will use boldface and parentheses because the original referenced item did.

Another handy use for `\oref` is in combination with `\nref` to refer to ranges:

```
The first two numbered examples in
this document were
were \oref[--\nref{atomism}]{atoms}.
```

The first two numbered examples in this document were (1–2).

`\lastref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

`\Lastref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

Produces a reference (with no hyperlink) to the most recently processed `\pitem` or `\ptag`, even without a `\label`. Useful for back-references in running text. With one argument, $\langle suffix \rangle$ is appended; with two, $\langle prefix \rangle$ is also prepended. Use `\lastref{}` for a plain reference. `\Lastref` is the titlecase variant.

`\nlastref`

`\nLastref`

Like `\lastref{}`, but returns the bare content without formatting. `\nLastref` is the titlecase variant.

`\parentref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

`\Parentref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as `\lastref`. `\Parentref` is the titlecase variant.

`\nparentref`

`\nParentref`

Like `\parentref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards. `\nParentref` is the titlecase variant.

`\parentref` and `\nparentref` are useful for making subitems whose names derive from their parent’s:

```

\DeclareNumberedStyle{inner}[
  counter format=\alph{inner},
  display format=#1.]
\begin{prop}
  \item[OI] Outer item.
  \label{outer2}
  \begin{prop}
    \item[style=inner,
      format=\parentref{.#1}]
    \label{dsub1}
    Ref: \ref{dsub1},
    naked: \nref{dsub1}.
    \item[style=inner,
      display format=#1.,
      ref format=\parentref{.#1}]
    \label{dsub2}
    Ref: \ref{dsub2},
    naked: \nref{dsub2}.
  \end{prop}
\end{prop}

```

OI Outer item.

OI.a Ref: **OI.a**, naked: a.

b. Ref: **OI.b**, naked: b.

The built-in `leveltwo` and `levelthree` styles have `ref format=\parentref{.#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as ‘(P1)’, a `leveltwo` sub-item references as ‘(P1a)’ and `\nref` returns just ‘a’.

7.1 How it works: `\propapply`

`\propapply{<template>}{<content>}`

Internally, each reference is stored in the `.aux` file as `\propapply{<template>}{<content>}`.

The `<template>` contains formatting with the placeholder `\propfmtarg` where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to `<content>`. The `\oref`^{P.11} and `\nref`^{P.11} commands work by locally redefining `\propapply`.

In normal use, you need not interact with `\propapply` directly.

`\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`.

8 Package options

`\proppositions{<keys>}`

Sets package-level keys. These can also be set:

- In the optional argument of `prop` and `inlineprop` (local to that environment).
- In the preamble with `\usepackage[<options>]{propositions}` (global).

Exception: `equations` is a global key; it cannot be used in the optional argument of `prop` or `inlineprop`.

8.1 List dimensions

`topsep`= $\langle length / length list \rangle$
`partopsep`= $\langle length / length list \rangle$
`itemsep`= $\langle length / length list \rangle$
`parsep`= $\langle length / length list \rangle$
`leftmargin`= $\langle length / length list \rangle$
`rightmargin`= $\langle length / length list \rangle$
`labelwidth`= $\langle length / length list \rangle$
`labelsep`= $\langle length / length list \rangle$
`itemindent`= $\langle length / length list \rangle$
`listparindent`= $\langle length / length list \rangle$

Override the standard L^AT_EX list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent`= $\langle length / length list \rangle$ (no default)

Positions the left edge of the label box at $\langle length \rangle$ from the enclosing margin, adjusting `\labelwidth`, `\leftmargin`, `\labelsep`, or `\itemindent` as needed. (Since the value of any one of these dimensions can be derived from those of the other four, it never makes sense to set all five.)

`tightspacing` (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (`topsep` and `itemsep` to 2pt with stretch/shrink, `parsep` to 0pt, `partopsep` to 1pt). Individual dimension keys set afterward override.

`nosep` (no value)

Sets `\topsep`, `\itemsep`, and `\parsep` all to zero.

`continue`= $\langle boolean \rangle$ (initially `true`)

When `true`, a `prop` environment that immediately follows another `prop` (with no intervening paragraph text) replaces the normal `\topsep`-based inter-list space with `\itemsep` + `\parsep`, giving the visual appearance of a single continuous list.

8.2 Default styles

When `\pitem` or `\ptag` is used without an explicit `style`, one of the following defaults is selected based on whether a `name/counter` was given and whether `ptag` mode is active.

`named style`= $\langle style \rangle$ (initially `proposition`)

The style used when a `name` or `counter` is given but no explicit `style`.

`named ptag style`= $\langle style \rangle$ (initially empty)

If set, overrides `named style` for `\ptag` items only.

`nameless style=<style>` (initially **numbered**)

The style used when `\pitem` or `\ptag` has no optional argument (no name, counter, or style).

`nameless ptag style=<style>` (initially **empty**)

If set, overrides `nameless style` for `\ptag` items only.

8.3 Environment-level style and item defaults

`style=<style>` (no default)

Sets a default style for all `\items` in this environment. The style's list-dimension keys (`leftmargin`, `labelwidth`, etc.) take effect immediately for the whole list; item-level keys are applied as defaults for each `\item`. An explicit `style=` on an individual `\item` overrides this default.

`items=<n>` (initially **1**)

Indicates that approximately $\langle n \rangle$ items are expected. Used only to compute `\propformlabel`: the preview label is formatted as if $\langle n \rangle$ items had already appeared. Has no effect on actual item processing.

`\propformlabel`

Expands to the formatted label of the first (or `items`-th) item expected in this environment, computed at `\begin{prop}`. Useful for setting list dimensions that depend on label width, e.g. `leftmargin=\widthof{\propformlabel}`. Updated after each `\item` to reflect the most recently output label.

8.4 Equation numbering

`equations` (no value, **global only**)

Sets `nameless style=eqnum` and installs hooks so that the L^AT_EX and `amstex` displayed equation environments (such as `equation` and `align`) use the same formatting as the `equation` prop style. The `equation` style's `display format` and `ref format` keys are used for generating equation numbers and cross-references to them. Redefining the `equation` style, e.g. with

```
| \SetPropStyle{equation}{format = [#1]}
```

will automatically update the hooks.

9 Compatibility

The `propositions` package is designed to work with `hyperref`, `cleveref`, and `amsmath`. `amsmath` is required for `\ptag` and the `equations` option. With `cleveref` loaded, all proposition items are assigned to a default proposition “reference type”; the `crefname` key can override this.

Recommended load order:

```
\usepackage{hyperref}  
\usepackage{amsmath} % if using \ptag  
\usepackage{propositions}  
\usepackage{cleveref} % if used
```

10 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an `ams-math` equation environment, `hyperref` may emit warnings of the form:

```
pdfTeX warning: destination with the same  
identifier (name{equation.N}) has been already  
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.